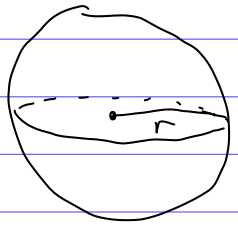
7



$V = V(t) =$

$r = r(t) =$

$\frac{dV}{dt} = -2$ Find $\frac{dr}{dt}$ when $2r = 10$ ← 2?

↓

$r = 5$

$V = \frac{4}{3}\pi r^3$

$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$

$-2 = \frac{4}{3} \cdot \pi \cdot 3 \cdot (5)^2 \frac{dr}{dt}$

$-2 = 4 \cdot \pi \cdot 25 \cdot \frac{dr}{dt}$

$-2 = 100\pi \cdot \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{-2}{100\pi} = \frac{-1}{50\pi} \approx -0.0006$

At the moment the diameter is 10 cm,
the radius shrinks at the rate of 0.0006 cm/min